Michael Liu

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Education

University of Waterloo *Sept 2025 - Apr 2030*
Bachelor of Computer Science - BCS, Computer Science

- Major Average: 93.1% | Cumulative Average: 92.7%
- Activities: Varsity Cross-Country, WAT.ai
- Courses: Object-Oriented Programming, Compilers and Low-Level Programming, Linear Algebra, Calculus, Functional Programming, Imperative Programming & Data Structures, Probability & Statistics

Job Experience

Machine Learning Research Assistant *Toronto, ON*
The Hospital for Sick Children (SickKids) *May 2026 - Aug 2026*

- Researched Reaction GFlowNets (RGFNs) for molecular generation, investigating sampling strategies for parallel synthesis.
- Designed and trained a simplified RGFN in Python to validate and study generative model behaviours.
- Implemented experiments in Python to evaluate sampling efficiency and learned policy behaviour.

Machine Learning Scientist *Waterloo, ON*
WAT.ai *Sept 2025 - Feb 2026*

- Collaborated with a small team of students on improving the segmentation of microaneurysms in fundus imaging.
- Co-authored a research paper detailing our methodology and results (GitHub).

Projects

CourtHawk: Tennis Computer Vision System *github/courthawk*
OpenCV, PyTorch, MediaPipe, Pandas, scikit-learn, FastAPI, TypeScript, React, Docker, Hugging Face

- Developed an end-to-end computer vision pipeline using YOLO, CNNs, and Pose Estimation to track players and tennis balls, detect court keypoints, and estimate player poses from the standard TV angle.
- Implemented shot detection and classification using player trajectories, ball tracking, and pose-derived features.
- Applied homography transformations to project player and ball positions onto a bird's-eye-view court, which are then used to analyze movement and shot speeds.
- Built and deployed a containerized full-stack application using Docker, FastAPI, React, and TypeScript, with pretrained models hosted on Hugging Face.

C++ Machine Learning Library from Scratch *github/cpp-ml-from-scratch*
C++, CMake

- Built a C++17 machine learning library from scratch using the C++ Standard Library centered around a tensor implementation, neural network layers, and classical machine learning algorithms.
- Designed a PyTorch-inspired interface with explicit training loops, backpropagation, and parameter updates to prioritize readability and transparency over abstraction.

Technical Skills

Programming Languages:	C/C++, Python, SQL, TypeScript
Data/Machine Learning:	PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, OpenCV
Databases/Tools:	SQL, Git, GitHub, LaTeX, Excel, Tableau, CMake, Docker, Hugging Face
Frontend:	React, HTML, CSS